

IRREGULARITIES IN SINGLE–CASE DELETION DIAGNOSTICS

D. R. Jensen

Virginia Polytechnic Institute

e-mail: djensen@vt.edu

Key words: Multiple outliers, influence diagnostics, induced noncentralities, inflated residuals.

Summary: Single–case deletion diagnostics rest on a single outlier at a point in the regressor space. Residuals are basic. Our objectives are to show (i) that shifts serve to inflate or deflate stochastically the ordinary squared residuals, even for non–outlying data; and (ii) that outliers generate anomalies as doubly noncentral distributions for both *outlier* and *influence* diagnostics, these specific to a given design. Outliers account for *masking* and *swamping*, with probabilities as evaluated here. Despite wide acceptance, software support, and routine useage, these anomalies despoil meanings historically ascribed to deletion diagnostics, thus abrogating objectives traditionally cited for their use. Case studies document some misdirected and unintended consequences of their continued use in practice.

1. Introduction

Regression diagnostics seek *leverages* of regressors, *outlying* data, observations *influential* in the analysis, and others. Texts include Belsley, Kuh and Welsch (1980), Cook and Weisberg (1982), Barnett and Lewis (1994), Atkinson (1985), Rousseeuw and Leroy (1987), Chatterjee and Hadi (1988), Myers (1990), and Fox (1991). Single–case deletions eliminate one row from the model; subset deletions eliminate $s > 1$ rows. Designs fully estimable after deletions are characterized in Ghosh (1978). Central to this study are statistics t_i and t_i^2 of Snedecor and Cochran (1968, p.157) for identifying a single shift at a designated locus in the regressor space. See also Beckman and Trussell (1974). The symbols are explained subsequently.

Finding a single outlier or influential observation is alleged to be “relatively simple from both analytical and computational points of view” Hadi and Simonoff (1993, p.1264); in part to explain the paucity of recent research. However, the outlying locus seldom is known, so as to require screening all of $\{t_i^2; 1 \leq i \leq n\}$; and “simplicity” seemingly rests on the premise that irregularities from a single shift vest exclusively in t_i^2 , values $\{t_j^2; j \neq i\}$ then being unaffected. This fails, along with “simplicity,” to reopen the matter afresh for the present study. Unaccounted are that shifts may occur *in lieu of*, or *in addition to*, the designated case Y_i , confuting *en mass* the historical trappings and reasons cited for their usage.

Our objectives are to show (i) that shifts may inflate or deflate stochastically the ordinary squared residuals, even for unshifted data; and (ii) that single and multiple outliers generate doubly noncentral distributions for designated diagnostics. The latter, as doubly noncentral t^2 , will be introduced in Section 2.2, and their parameter values evaluated numerically in Sections 4 and 5 for specified designs. An outline follows.

Section 2 surveys notation, distributions, models and their errors, and a dichotomy of projected shifts. Section 3 establishes irregularities in the squared residuals, to be inflated or deflated stochastically by outliers at other loci, with consequent implications regarding *robust regression*. Irregularities in $\{t_i^2; 1 \leq i \leq n\}$ and related diagnostics are examined, to include unequivocal venues for the *masking* and *swamping* of outliers. These findings are illustrated using informative small examples in Section 4, together with a clarification in Section 5 of more extensive findings from the literature. Essentials are summarized in Section 6; further supporting facts are given in Appendix A.

2. Preliminaries

2.1. Notation

Here $\mathbb{R}^n$ designates Euclidean n -space, $\mathbb{R}_+^n$ its positive orthant, and $\mathbb{S}_n$ the real symmetric $(n \times n)$ matrices. Matrices and vectors are in bold type; the transpose and inverse of $\mathbf{A}$ are $\mathbf{A}'$ and $\mathbf{A}^{-1}$, whereas $\mathbf{I}_n$ is the $(n \times n)$ identity and $\text{Diag}(a_1, \dots, a_k)$ a diagonal array. Given $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_k]$ of order $(n \times k)$ and rank $k < n$, its *column span*, $S_p(\mathbf{B})$, is the k -dimensional proper subspace of $\mathbb{R}^n$ spanned by $[\mathbf{b}_1, \dots, \mathbf{b}_k]$. The eigenvalues of $\mathbf{A} \in \mathbb{S}_n$ are $\{\lambda_i(\mathbf{A}) = \alpha_i; 1 \leq i \leq n\}$, ordered as $\{\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_n\}$; its spectral decomposition is $\mathbf{A} = \mathbf{P}\mathbf{D}_\alpha\mathbf{P}'$ with $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_n]$ orthogonal and $\mathbf{D}_\alpha = \text{Diag}(\alpha_1, \dots, \alpha_n)$. For $\{\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_r > 0 = \alpha_{r+1} = \dots = \alpha_n\}$, partition $\mathbf{P} = [\mathbf{P}_1, \mathbf{P}_2]$ conformably, with $\mathbf{P}_1 = [\mathbf{p}_1, \dots, \mathbf{p}_r]$ and $\mathbf{P}_2 = [\mathbf{p}_{r+1}, \dots, \mathbf{p}_n]$. Then the *range* and *null* spaces of $\mathbf{A}$ are respectively $\mathcal{R}(\mathbf{A}) = S_p(\mathbf{P}_1)$ and $\mathcal{N}(\mathbf{A}) = S_p(\mathbf{P}_2)$. Any *g-inverse* $\mathbf{A}^-$ of $\mathbf{A}$ meets $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$.

2.2. Special distributions

Identify by $\mathcal{L}(\mathbf{Y})$ the distribution of $\mathbf{Y} \in \mathbb{R}^n$, and by $\mathbf{E}(\mathbf{Y})$ and $\mathbf{V}(\mathbf{Y})$ its mean vector and dispersion matrix, taking $\text{Var}(Y) = \sigma^2$ on $\mathbb{R}^1$. Specifically, $\mathcal{L}(\mathbf{Y}) = N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is Gaussian on $\mathbb{R}^n$ with $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ as its mean and dispersion parameters. The cumulative distribution function (*cdf*) of a random $U \in \mathbb{R}^1$ is $F_U(u)$. Distributions of note on $\mathbb{R}_+^1$ include $\mathcal{L}(U^2) = \chi^2(\nu, \lambda)$ as chi-squared having ν degrees of freedom and noncentrality $\lambda \geq 0$, whereas $\mathcal{L}(|U|) = \chi(\nu, \lambda)$ is the noncentral Rayleigh distribution. Under scaling $U^2 \rightarrow W^2 = c^2 U^2$, we designate $\mathcal{L}(W^2) = c^2 \chi^2(\nu, \lambda)$ and $\mathcal{L}(|W|) = |c| \chi(\nu, \lambda)$. In addition, taking $\mathcal{L}(U^2) = \chi^2(1, \lambda_1)$ and $\mathcal{L}(V^2) = \chi^2(\nu, \lambda_2)$ to be distributed independently, the distribution of $t_0^2 = \nu U^2 / V^2$, namely $\mathcal{L}(t_0^2) = t^2(\nu, \lambda_1, \lambda_2)$, is Student's doubly noncentral t^2 -distribution. Moreover, $\{t_0^2 > c_\alpha^2\}$ is the conventional α -level rejection rule against two-sided alternatives, based on the central distribution $t^2(\nu, 0, 0)$. Recall that $t^2(\nu, \lambda_1, \lambda_2)$ increases stochastically with increasing λ_1 , and decreases stochastically with increasing λ_2 , other parameters held fixed.

Remark 1 Taking $\mathbf{E}(e_i) = 0$ and $\mathcal{L}(S_i^{*2}) = \chi^2(\nu, \lambda)$ with $\lambda > 0$, the doubly noncentral t -distributions derive here from $\mathcal{L}(t_i^*) = \mathcal{L}((e_i + \delta)/(S_i^* \sqrt{1 - h_{ii}}))$, to be designated as $\mathcal{L}(t_i^*) = t(\nu; \delta, \lambda)$, where

δ itself may be positive or negative. For further details see Nandi and Choudhury (2002) and references cited.

2.3. The models

A response Y_j relates to regressors $\{X_{j1}, \dots, X_{jk}\}$ through $p = k + 1$ parameters $\boldsymbol{\beta}' = [\beta_0, \beta_1, \dots, \beta_k]$ as $\{Y_j = \beta_0 + \beta_1 X_{j1} + \dots + \beta_k X_{jk} + \varepsilon_j; 1 \leq j \leq n\}$. Arrayed as $\mathbf{Y}_0 = \mathbf{X}_0 \boldsymbol{\beta} + \boldsymbol{\varepsilon}_0$, for computational consistency we rearrange these as

$$\begin{bmatrix} \mathbf{Y} \\ Y_i \end{bmatrix} = \begin{bmatrix} \mathbf{X} \\ \mathbf{x}_i' \end{bmatrix} \boldsymbol{\beta} + \begin{bmatrix} \boldsymbol{\varepsilon} \\ \varepsilon_i \end{bmatrix}, \quad (1)$$

where $[Y_i, \mathbf{x}_i', \varepsilon_i]$ is the row to be deleted. Let $\mathbf{H} = [h_{ij}] = \mathbf{X}_0(\mathbf{X}_0' \mathbf{X}_0)^{-1} \mathbf{X}_0'$; its diagonal elements $\{h_{ii}; 1 \leq i \leq n\}$ are *leverages* attributed to rows $\{\mathbf{x}_i'; 1 \leq i \leq n\}$ of $\mathbf{X}_0$, whereas $\{h_{ij}; i \neq j\}$ are *coleverages*. Then the residuals from the full model comprise $\mathbf{e}_0 = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}_0$, to be partitioned as $\mathbf{e}_0' = [\mathbf{e}', e_i]$ conformably with (1). The *OLS* and *MSE* estimators are $(\hat{\boldsymbol{\beta}}, S^2)$ and $(\hat{\boldsymbol{\beta}}, S_i^2)$ from the full and reduced data, respectively, where $\hat{\boldsymbol{\beta}}' = [\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k]$ and $\hat{\boldsymbol{\beta}}_i' = [\hat{\beta}_{i0}, \hat{\beta}_{i1}, \dots, \hat{\beta}_{ik}]$. For subset deletions the single row $[Y_i, \mathbf{x}_i', \varepsilon_i]$ in (1) is replaced by $[\mathbf{Y}_I, \mathbf{X}_I, \boldsymbol{\varepsilon}_I]$ as the $s > 1$ rows to be deleted, taking F_I of Gentleman and Wilk (1975) to track a vector shift at $s > 1$ deleted observations.

Definition 1 Principal diagnostics are $t_i^2 = e_i^2 / S_i^2 (1 - h_{ii})$ and $F_I = \mathbf{e}_I' (\mathbf{I}_s - \mathbf{H}_{II})^{-1} \mathbf{e}_I / s S_I^2$, with $\mathbf{e}_0' = [\mathbf{e}', e_i]$ and $(\mathbf{I}_s - \mathbf{H}_{II})$ as the lower right $(s \times s)$ block of the partitioned matrix $(\mathbf{I}_n - \mathbf{H})$, and S_I^2 as the *MSE* from the reduced data on deleting $[\mathbf{Y}_I, \mathbf{X}_I, \boldsymbol{\varepsilon}_I]$.

Assumptions next pertain to moments and distributions of errors. Conventional models for shifted outliers take $\{Y_i \rightarrow Y_i + \delta_i\}$ exclusively at the deleted observation, and shifts $\boldsymbol{\gamma} = \mathbf{0}$ elsewhere. To the contrary, here unfettered shifts $\{\mathbf{Y} \rightarrow \mathbf{Y} + \boldsymbol{\gamma}\}$ are allowed among the observations not deleted, where $\{\mathbf{Y}_0 \rightarrow \mathbf{Y}_0 + \boldsymbol{\omega}\}$ with $\boldsymbol{\omega}' = [\boldsymbol{\gamma}', \delta]$; specifically $[\mathbf{Y}', Y_i] \rightarrow [(\mathbf{Y} + \boldsymbol{\gamma})', (Y_i + \delta)]$.

Assumptions A.

A1. $E(\boldsymbol{\varepsilon}) = \boldsymbol{\gamma} \in \mathbb{R}^r$ and $E(\varepsilon_i) = \delta \in \mathbb{R}^1$;

A2. $V(\boldsymbol{\varepsilon}_0) = \sigma^2 \mathbf{I}_n$; and

A3. $\mathcal{L}([\boldsymbol{\varepsilon} - \boldsymbol{\gamma}', (\varepsilon_i - \delta)]) = N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Recall the matrices $\mathbf{H}_r = \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'$ and $(\mathbf{I}_r - \mathbf{H}_r)$ to be idempotent, projecting into the “Regressors” and “Error” spaces generated by $\{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}\}$, with $\mathbf{Y}$ of order $(r \times 1)$. This supports the following classification for shifts, with consequences seen subsequently in regard to noncentral parameters induced through shifts.

Definition 2 Consider $\{\mathbf{Y} \rightarrow \mathbf{Y} + \boldsymbol{\gamma}\}$ in reference to $\{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}\}$, so that $E(\boldsymbol{\varepsilon}) = \boldsymbol{\gamma}$ as in Assumption A.1. Then $\boldsymbol{\gamma} \in \mathbb{R}^r$ is decomposed as $\boldsymbol{\gamma} = \boldsymbol{\gamma}_1 + \boldsymbol{\gamma}_2$, with $\boldsymbol{\gamma}_1 = \mathbf{H}_r \boldsymbol{\gamma} \in \mathcal{R}(\mathbf{H}_r)$ and $\boldsymbol{\gamma}_2 = (\mathbf{I}_r - \mathbf{H}_r) \boldsymbol{\gamma} \in \mathcal{N}(\mathbf{H}_r)$.

To continue, both leverages, as diagonal elements of $\mathbf{H}_n = \mathbf{X}_0(\mathbf{X}_0' \mathbf{X}_0)^{-1} \mathbf{X}_0'$, and coleverages as off-diagonal elements, are informative. Welsch (1983), through derivatives, identifies a point of high leverage as “an observation whose fitted value is largely determined by the corresponding response value.” In parallel with Jensen (2010) is the following.

Remark 2 The mixed derivatives $\{\partial \hat{Y}_i / \partial Y_j = h_{ij}\}$ identify a pair (Y_i, Y_j) with high leverage $|h_{ij}|$ as “an observation Y_i whose fitted value $\hat{Y}_i$ is increasing or decreasing in Y_j at the rate $|h_{ij}|$.”

Definition 3 Critical to users are the *masking* and *swamping* of outliers: *Masking* when outliers are hidden; *Swamping* when non-outliers are misidentified due to outliers elsewhere.

3. The principal findings

3.1. Overview

A single shift $\{Y_i \rightarrow Y_i + \delta_i\}$ accounts for the stochastic inflation of $e_i^2 = (Y_i - \hat{Y}_i)^2$, as tacitly acknowledged in the literature. This is extended here to include $e_j^2 = (Y_j - \hat{Y}_j)^2$ owing to shifts at other loci, despite that Y_j remains unshifted. This is undertaken in Section 3.2. Moreover, shifts $\{\mathbf{Y}_0 \rightarrow \mathbf{Y}_0 + \boldsymbol{\omega}\}$ induce $\{\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2); 1 \leq i \leq n\}$ as doubly noncentral, where $v = n - p - 1$. Specializing identifies (i) nonzero shifts $\{\delta_i \neq 0\}$ that remain undetected; (ii) values $\boldsymbol{\gamma} \neq \mathbf{0}$ nonetheless preserving $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, 0)$ for some $\lambda_1 > 0$; (iii) the occurrence of “regression effects” among shifts and their consequences; and (iv) that outliers may either depress or inflate t_i^2 stochastically; see Section 3.3. In short, anomalies persist.

3.2. Inflated residuals

Under a single shift $\{Y_i \rightarrow Y_i + \delta_i\}$, $e_i^2 = (Y_i - \hat{Y}_i)^2$ is inflated stochastically from equation (A.3) of Jensen and Ramirez (2014), since $\mathcal{L}(e_i^2 / \sigma^2(1 - h_{ii})) = \chi^2(1, \delta_i^2)$. However, distributions of the $n - 1$ other values $\{e_j^2; j \neq i\}$ are not necessarily central, even when observations $\{Y_j; j \neq i\}$ remain unshifted. Nor does it follow for shifted Y_i that e_i^2 necessarily is inflated stochastically, but may be deflated when allowance is made for shifts at other loci. These are considered next for squared and absolute values of the *OLS* residuals.

Theorem 1 Under Gaussian errors with $\sigma^2 = 1.0$, consider $\{e_i^2 = (Y_i - \hat{Y}_i)^2; 1 \leq i \leq n\}$ and their moduli $\{|e_i|; 1 \leq i \leq n\}$, with $\lambda_i = (\mathbf{g}_{21}\boldsymbol{\gamma}_1 + g_{22}\delta_i)'g_{22}^{-1}(\mathbf{g}_{21}\boldsymbol{\gamma}_1 + g_{22}\delta_i)$ as in Theorem 2(i).

- (i) Given the single shift $\{Y_i + \delta_i\}$, $\mathcal{L}(e_i^2)$ is scaled noncentral chi-squared, specifically $\mathcal{L}(e_i^2) = (1 - h_{ii})\chi^2(1, \lambda_i)$, having $E(e_i^2) = (1 - h_{ii})(1 + \lambda_i)$, $\text{Var}(e_i^2) = (1 - h_{ii})^2 2(1 + 2\lambda_i)$, and third central moment $M_3 = (1 - h_{ii})^3 8(1 + 3\lambda_i)$.
- (ii) $\mathcal{L}(|e_i|)$ is scaled noncentral Rayleigh, namely $\mathcal{L}(|e_i|) = (1 - h_{ii})^{\frac{1}{2}}\chi(1, \lambda_i)$, having expectation

$$E(|e_i|) = (1 - h_{ii})^{\frac{1}{2}} \frac{[(1 + \lambda_i)^2 - \frac{1}{4}(1 + 2\lambda_i)]}{(1 + \lambda_i)^{\frac{3}{2}}} + O(M_3) \quad (2)$$

and, at $\lambda_i = 0$, then $E(|e_i|) = 3(1 - h_{ii})^{\frac{1}{2}}/4 + O(M_3)$.

- (iii) Suppose $\delta_i \neq 0$ yet the “regression effect” $\{\delta_i = \mathbf{c}_i'\boldsymbol{\gamma}\}$ holds with $\mathbf{c}_i' = -\mathbf{g}_{21}/g_{22}$. Then distributions $\{\mathcal{L}(e_i^2), \mathcal{L}(|e_i|)\}$ are central, masking through residuals that $\{Y_i \rightarrow Y_i + \delta_i\}$ has shifted.

Proof. That $\mathcal{L}(e_i^2) = (1 - h_{ii})\chi^2(1, \lambda_i)$ follows from Lemma A.4(i) of Jensen and Ramirez (2014). Moments follow routinely on scaling as in conclusion (i). Moreover, $\mathcal{L}(|e_i|/\sigma\sqrt{1 - h_{ii}}) = \chi(1, \lambda_i)$ follows in support of conclusion (ii). To continue, let $Z > 0$ have mean μ , variance σ^2 , and third central moment M_3 . The first three terms of the Taylor expansion for $U = \sqrt{Z}$ give

$$E(U) = \sqrt{\mu} + 0 - (1/8)\mu^{-\frac{3}{2}}\sigma^2 + O(E(Z - \mu)^3). \quad (3)$$

A direct application gives $E(|e_i|) \doteq (1 - h_{ii})^{\frac{1}{2}}[(1 + \lambda_i)^2 - \frac{1}{4}(1 + 2\lambda_i)]/(1 + \lambda_i)^{\frac{3}{2}}$ to the order of approximation in (3). Conclusion (iii) follows as in Theorem 3(v). ■

3.3. Induced noncentralities

Let $\mathbf{G} = (\mathbf{I}_n - \mathbf{H}_n)$ as before; the residual vector $\mathbf{e}_0 = \mathbf{G}\mathbf{Y}_0$ is now partitioned as

$$\begin{bmatrix} \mathbf{e} \\ e_i \end{bmatrix} = \begin{bmatrix} (\mathbf{I}_r - \mathbf{H}_{00}) & -\mathbf{h}_{0i} \\ -\mathbf{h}_{i0} & (1 - h_{ii}) \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ Y_i \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{11} & \mathbf{g}_{12} \\ \mathbf{g}_{21} & g_{22} \end{bmatrix} \begin{bmatrix} \mathbf{Y} \\ Y_i \end{bmatrix}, \quad (4)$$

where $\mathbf{H}_{00} = \mathbf{X}(\mathbf{X}_0'\mathbf{X}_0)^{-1}\mathbf{X}'$; $\mathbf{h}_{0i} = \mathbf{X}(\mathbf{X}_0'\mathbf{X}_0)^{-1}\mathbf{x}_i$; $h_{ii} = \mathbf{x}_i'(\mathbf{X}_0'\mathbf{X}_0)^{-1}\mathbf{x}_i$ is the leverage at $\mathbf{x}_i$; and $\mathbf{h}_{i0} = \mathbf{h}_{0i}'$ comprise coleverages between $[\mathbf{x}_1, \dots, \mathbf{x}_r]$ and $\mathbf{x}_i$. The following is basic.

Theorem 2 Given Assumptions A, such that $E(\mathbf{e}_0) = \boldsymbol{\omega} = [\boldsymbol{\gamma}', \delta]'$; decompose $\boldsymbol{\gamma} = \boldsymbol{\gamma}_1 + \boldsymbol{\gamma}_2$ as in Definition 2; and let $v = n - p - 1$. Then the distribution of t_i^2 is the doubly noncentral $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2)$ with

- (i) $\lambda_1 = (\mathbf{g}_{21}\boldsymbol{\gamma}_1 + g_{22}\delta)'g_{22}^{-1}(\mathbf{g}_{21}\boldsymbol{\gamma}_1 + g_{22}\delta)$;
- (ii) $\lambda_2 = \boldsymbol{\gamma}_2'(\mathbf{I}_r - \mathbf{H}_r)\boldsymbol{\gamma}_2$.

Proof. These follow on specializing Theorem 1 of Jensen and Ramirez (2014) to $s = 1$. ■

For shifts $\boldsymbol{\gamma} \neq \mathbf{0}$, distributions of t_i^2 emerge as doubly noncentral, or as singly noncentral in the numerator or denominator. The decomposition $\boldsymbol{\gamma} = \boldsymbol{\gamma}_1 + \boldsymbol{\gamma}_2$ of Definition 2 is basic; it effects the conversion of shifts into noncentral parameters as follows, on recalling that $\boldsymbol{\gamma}_1 = \mathbf{H}_r\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{H}_r)$ and $\boldsymbol{\gamma}_2 = (\mathbf{I}_r - \mathbf{H}_r)\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{I}_r - \mathbf{H}_r) = \mathcal{N}(\mathbf{H}_r)$. Generally speaking, $\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{H}_r)$ accounts for $t^2(v, \lambda_1, 0)$, whereas $\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{I}_r - \mathbf{H}_r)$ gives $t^2(v, \lambda_1, \lambda_2)$ with $\lambda_2 > 0$. Details follow, where $\mathbf{G}_{121} = \mathbf{g}_{12}g_{22}^{-1}\mathbf{g}_{21}$ is of order $(r \times r)$ and unit rank.

Theorem 3 Consider $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2)$ as in Theorem 2 under Assumptions A. Essential properties under further structure emerge as follow.

- (i) If $\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{H}_r)$, then $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, 0)$ with λ_1 as in Theorem 1, and at $\delta = 0$, then $\lambda_1 = \boldsymbol{\gamma}'\mathbf{G}_{121}\boldsymbol{\gamma} > 0$.
- (ii) If $\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{I}_r - \mathbf{H}_r)$, then $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2)$ with $\lambda_1 = (1 - h_{ii})\delta^2$ and $\lambda_2 = \boldsymbol{\gamma}'(\mathbf{I}_r - \mathbf{H}_r)\boldsymbol{\gamma} > 0$; and at $\delta = 0$, $\mathcal{L}(t_i^2) = t^2(v, 0, \lambda_2)$.
- (iii) If $\delta = 0$ and $\boldsymbol{\gamma} \in \mathcal{R}(\mathbf{G}_{121})$, then $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, 0)$ with $\lambda_1 = \boldsymbol{\gamma}'\mathbf{G}_{121}\boldsymbol{\gamma} > 0$; and
- (iv) If $\delta = 0$ and $\boldsymbol{\gamma} \in \mathcal{N}(\mathbf{G}_{121})$, then $\mathcal{L}(t_i^2) = t^2(v, 0, \lambda_2)$ with $\lambda_2 = \boldsymbol{\gamma}'(\mathbf{I}_r - \mathbf{H}_r)\boldsymbol{\gamma} > 0$.

(v) For $\gamma \in \mathcal{R}(\mathbf{H}_r)$, the “Regression Effect” $\delta = \mathbf{c}'\gamma$ with $\mathbf{c}' = -\mathbf{g}_{21}/g_{22}$, yields $\lambda_1 = 0$.

Proof. These follow on specializing to $s = 1$ Theorem 2 of Jensen and Ramirez (2014). ■

The foregoing results encompass all diagnostics $\{Dg_i \in \mathfrak{D}\}$ as listed in Appendix A. A complete accounting for noncentral distributions induced by shifts is given next.

Corollary 1 Let $Dg_i \in \mathfrak{D}$ be a typical deletion diagnostic. Then representations for $\mathcal{L}(Dg_i)$ are given generally as $\mathcal{L}(Dg_i) = \mathcal{L}[\psi(t_i)]$, with $\psi(t_i)$ as the mapping from t_i to an element of $\mathfrak{D}$. Specifically, $\mathcal{L}(t_i^2) = t^2(v; \lambda_1, \lambda_2)$ under shifts as in Theorems 2 and 3, and $\mathcal{L}(t_i) = t(v, \tau, \lambda_2)$, where $\tau = \pm\sqrt{\lambda_1}$, emerge as appropriate to one-sided upper or lower shifts.

Proof. Using different methods, Lamotte (1999) and Jensen (2000) show the mappings $\psi(t_i)$ from t_i to $\mathfrak{D}$ to be one-to-one in t_i or t_i^2 . Expressions for $\psi(t_i)$ are derived algebraically and listed in Jensen (2000), regardless of the distribution of t_i . That $\mathcal{L}(Dg_i) = \mathcal{L}[\psi(t_i)]$ for each diagnostic now follows from elementary change-of-variables in distribution theory, either from $t^2(v, \lambda_1, \lambda_2)$ as in Theorems 2 and 3, or from $t(v, \tau, \lambda_2)$ with $\tau = \pm\sqrt{\lambda_1}$ as in Remark 1 for one-sided upper or lower shifts, to complete our proof. ■

Remark 3 Recall that $t^2(v, \lambda_1, \lambda_2)$ increases stochastically with increasing λ_1 , and decreases with increasing λ_2 . Nonetheless, a given $t^2(v, \lambda_1, \lambda_2)$ may be compared against $t^2(v, 0, 0)$ at some argument c^* of their *cdfs* as reference. This follows on computing $\alpha^* = P(t^2 > c^*)$ from $\mathcal{L}(t^2) = t^2(v, 0, 0)$, then $\beta = P(t_i^2 > c^*)$ from $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2)$. Then $\beta < \alpha^*$ identifies $\mathcal{L}(t_i^2)$ as skewed to the left of $t^2(v, 0, 0)$ at c^* , whereas $\beta > \alpha^*$ identifies $\mathcal{L}(t_i^2) = t^2(v, \lambda_1, \lambda_2)$ as skewed to its right. Further usage is explained later in context.

3.4. Anomalies: A critique

Irregularities in $\{\mathcal{L}(|e_i|), \mathcal{L}(e_i^2), \mathcal{L}(t_i^2); 1 \leq i \leq n\}$ bear heavily on topics in inference. Some consequences are listed next; others may be drawn by the reader. Issues persist in the *masking* and *swamping* of outliers, both “important . . . in influence analysis which deserves further study;” Hoaglin and Kempthorne (1986, p.410). Letting $v = n - p - s$, we have the following.

Properties.

- P1(i)** Testing $H_0 : \delta_i = 0$ vs $H_1 : \delta_i \neq 0$ using $\{t_i^2 > c_\alpha^2\}$ is exact at level α when $\omega' = [\mathbf{0}', \delta_i]$.
- (ii) For $\gamma \in \mathcal{N}(\mathbf{H}_r)$ the test is conservative with level $\alpha^* < \alpha$ since $t^2(v, 0, \lambda_2)$ is stochastically smaller than the nominal $t^2(v, 0, 0)$ from Theorem 3(ii).
- (iii) If $\delta_i \approx \mathbf{c}'\gamma$ holds approximately in Theorem 3(v), so that λ_1 is small and $\{t_i^2 < c_\alpha^2\}$, then the mistaken inference that $\delta_i = 0$ would serve to *mask* $\delta_i \neq 0$.
- (iv) For $\gamma \in \mathcal{R}(\mathbf{G}_{121})$, a “significant” outcome $\{t_i^2 > c_\alpha^2\}$, misattributed to $H_1 : \delta_i \neq 0$ when in fact $\delta_i = 0$, effects a venue for $\gamma \neq \mathbf{0}$ to *swamp* $\delta_i = 0$ from Theorem 3(iii).
- P2(i)** Large, intermediate, or small t_i^2 may attribute to outliers: Large values for $\lambda_1 > 0$ in Theorem 3(i)(iii); small values for $\lambda_2 > 0$ in Theorem 3(iv) and Theorem 3(ii) at $\delta = 0$; and intermediate values for λ_1 compensating λ_2 in Theorem 2 and Theorem 3(ii).

- (ii) These serve to mask both $\delta \neq 0$ and $\gamma \neq \mathbf{0}$.

These issues are best exemplified numerically. Case studies to these ends are undertaken next.

4. Case A: Quadratic response

Noncentralities for a prospective design may be assessed beforehand. Case studies, first for a small transparent data set, then for examples from the literature, illustrate the scope of the irregularities. Computations utilize Minitab, PROC IML of the S A S Programming System, and Maple.

4.1. The model

Take $\{Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i; 1 \leq i \leq 5\}$ and design points from a Central Composite Design (CCD) of Box and Wilson (1951), where $\{(Y_i, X_i); 1 \leq i \leq 5\}$ fill the first two rows of Table 1; $\mathbf{x}_i' = [1, X_i, X_i^2]$

Table 1: Values $\{(Y_i, X_i, h_{ii}); 1 \leq i \leq 5\}$ and $[h_{50}, h_{55}]$ for a single regressor CCD design under a quadratic model.

i	1	2	3	4	5
Y_i	-2.00020	6.13322	10.86978	10.15305	9.48890
X_i	-1.73205	-1.00000	0.00000	1.00000	1.73205
h_{ii}	0.84722	0.37500	0.55555	0.37500	0.84722
$[h_{50}, h_{55}]$	0.09722	-0.13317	-0.11111	0.29984	0.84722

are typical rows of $\mathbf{X}_0(5 \times 3)$; diagonal elements of $\mathbf{H}_5 = \mathbf{X}_0(\mathbf{X}_0' \mathbf{X}_0)^{-1} \mathbf{X}_0'$ are leverages listed in the third row; and the fourth row coincides with the final row of $\mathbf{H}_5$, namely $[h_{51}, h_{52}, h_{53}, h_{54}, h_{55}] = [h_{50}', h_{55}]$ comprising coleverages with and the leverage of $\mathbf{x}_5$. Since $n - p - 1 = 1$, the reference of Remark 3 is taken as the 75th percentile, $c^* = 5.8285$, of $t^2(1, 0, 0)$ owing to its excessive upper tail.

Applications of various findings are immediate: (i) The extremal coleverage of $[\mathbf{x}_1, \dots, \mathbf{x}_4]$ with $\mathbf{x}_5$ is 0.29984 from Table 1, so that Y_4 exerts the greatest impact on $\hat{Y}_5$, a function increasing at rate $d\hat{Y}_5/dY_4 = 0.29984$ as in Remark 2. (ii) Values for the regression effects $\{\delta_i = \mathbf{c}_i' \gamma\}$ of Theorem 3(v), evaluated at $i = 5$, are $\mathbf{c}_5' = [0.63634, -0.87165, -0.72726, 1.96256]$ from the final row of Table 1. (iii) The leverage h_{33} at $X_3 = 0.000$, the precise center, exceeds both h_{22} and h_{44} , refuting assertions that points of high leverage are remote in the regressor space, and conversely; see Belsley et al. (1980, pp16–17); Chatterjee and Hadi (1988, p.95); Hocking and Pendleton (1983, p.511); Myers (1990, p.252); and Rousseeuw and van Zomeren (1990, p.634), for example. While true for multilinear models, the claim clearly is model-dependent.

4.2. Single outliers

In pursuit of shifts, convention would deploy $\{t_1^2, t_2^2, t_3^2, t_4^2, t_5^2\}$ under successive single-case deletions. Noncentrality from a single shift $\{Y_i \rightarrow Y_i + \delta_i\}$ does not vest exclusively in t_i^2 as noted, but

distributes instead across $\{t_j^2; j \neq i\}$. Details are examined first for a single shift, to be expanded subsequently.

Case Study 1. To consider effects on $\mathcal{L}(t_5^2 | \delta_j)$ induced by shifts $\{\delta_j; 1 \leq j \leq 4\}$ at preceding loci, deleting row $\mathbf{x}_5'$ from $\mathbf{X}_0$ gives $\mathbf{H}_4 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and

$$(\mathbf{I}_4 - \mathbf{H}_4) = \begin{bmatrix} 0.090909 & -0.215090 & 0.181818 & -0.057630 \\ -0.215090 & 0.508916 & -0.430190 & 0.136364 \\ 0.181818 & -0.430190 & 0.363636 & -0.115270 \\ -0.057630 & 0.136364 & -0.115270 & 0.036539 \end{bmatrix}. \quad (5)$$

To examine effects on $\{\mathcal{L}(t_5^2 | \delta_j); 1 \leq j \leq 4\}$, take $\{Y_j + \delta_j; 1 \leq j \leq 4\}$ in succession with $\delta_5 = 0$. For the case $\mathcal{L}(t_5^2 | \delta_1)$, Theorem 2 applies with $\boldsymbol{\omega}' = [\delta_1, 0, 0, 0, 0]$ and $\boldsymbol{\gamma}' = [\delta_1, 0, 0, 0]$ where δ of Theorem 2, now δ_5 , is zero since Y_5 remains unshifted. We recover $\lambda_{1i} = \boldsymbol{\gamma}' \mathbf{g}_{12} \mathbf{g}_{22}^{-1} \mathbf{g}_{21} \boldsymbol{\gamma}$ at $i = 1$ as in Theorem 2(i), with $\mathbf{g}_{21} = -\mathbf{h}_{50}$ and $g_{22} = (1 - h_{55})$ from Table 1. The $(1, 1)$ element of $\mathbf{g}_{12} \mathbf{g}_{22}^{-1} \mathbf{g}_{21}$, of order (4×4) , is $0.097222/0.15728 = 0.06187$, giving $\lambda_{11} = 0.06187 \delta_1^2$ as in Table 2 under $\mathcal{L}(t_5^2 |$

Table 2: Noncentralities in $\mathcal{L}(t_5^2 | \delta_i) = t^2(1, \lambda_{1i}, \lambda_{2i})$ induced in succession by $\{Y_i + \delta_i; 1 \leq i \leq 4\}$ when $\delta_5 = 0$, together with probabilities $\beta(\delta)$ with $\delta = \pm 2$.

Noncentralities	$Y_1 \rightarrow Y_1 + \delta_1$	$Y_2 \rightarrow Y_2 + \delta_2$	$Y_3 \rightarrow Y_3 + \delta_3$	$Y_4 \rightarrow Y_4 + \delta_4$
λ_{1i}	$0.06187 \delta_1^2$	$0.11608 \delta_2^2$	$0.08081 \delta_3^2$	$0.58845 \delta_4^2$
λ_{2i}	$0.09091 \delta_1^2$	$0.50892 \delta_2^2$	$0.36364 \delta_3^2$	$0.03654 \delta_4^2$
$\beta(\pm 2)$	0.23614	0.12443	0.15102	0.43524

$\delta_1)$. For $\mathcal{L}(t_5^2 | \delta_2)$ we have $\boldsymbol{\omega}' = [0, \delta_2, 0, 0, 0]$ and $\boldsymbol{\gamma}' = [0, \delta_2, 0, 0]$. The $(2, 2)$ element of $\mathbf{g}_{12} \mathbf{g}_{22}^{-1} \mathbf{g}_{21}$ is $h_{52}^2/(1 - h_{55}) = (-0.13317)^2/0.15728 = 0.11608$ from Table 1, giving $\lambda_{12} = 0.11608 \delta_2^2$ as in Table 2. Values for $\{\lambda_{13}, \lambda_{14}\}$ follow similarly. Denominator noncentralities from Theorem 2(ii) are $\{\lambda_{2i} = \boldsymbol{\gamma}_i' (\mathbf{I}_4 - \mathbf{H}_4) \boldsymbol{\gamma}_i; 1 \leq i \leq 4\}$, namely, successive products of diagonal elements of (5) by δ_i^2 , as reported in Table 2. The listed probabilities are $P(t_i^2 > c^*)$ with $c^* = 5.8285$ as reference.

Case Study 2. Again consider $\{t_1^2, t_2^2, t_3^2, t_4^2, t_5^2\}$ under single shifts, but now take $\{Y_1 \rightarrow Y_1 + \delta_1\}$ exclusively, others to remain unshifted. Parameters for $\mathcal{L}(t_i^2 | \delta_1) = t^2(1, \lambda_{11}, \lambda_{12})$ are listed for $\{1 \leq i \leq 5\}$ in columns 2 and 3 of Table 3.

Computations proceed as before. The reader is reminded that for t_1^2 , the case $ID = (2345 \cdot 1)$ entails reordering the rows of $\mathbf{X}_0$ in keeping with expression (1). In particular, computations for $\mathcal{L}(t_2^2 | \delta_1)$ use the squared leverage, scaled in turn by $(1 - h_{22})$, found to be $h_{21}^2/(1 - h_{22}) = (0.29984)^2/(1 - 0.3750) = 0.14385$, giving $\lambda_{11} = 0.14385 \delta_1^2$ for t_2^2 as in row 2, column 2 of Table 3. On deleting $\mathbf{x}_2$ and recomputing $\mathbf{H}_4$, Theorem 2(ii) now gives $\lambda_{12} = 0.00893 \delta_1^2$ for t_2^2 in Table 3. Parallel computations complete columns 2 and 3 of Table 3.

In contrast to leverage 0.84722 at $\mathbf{x}_1$, consider instead $\{Y_2 \rightarrow Y_2 + \delta_2\}$ as a single shift with leverage 0.37500. We have $\{\mathcal{L}(t_i^2 | \delta_2) = t^2(1, \lambda_{21}, \lambda_{22}); 1 \leq i \leq 5\}$, listing $(\lambda_{21}, \lambda_{22})$ as columns 5 and 6 of Table 3. Computations proceed as before.

A Critique. Unforeseen irregularities inject onto $\{t_1^2, t_2^2, t_3^2, t_4^2, t_5^2\}$ through single shifts; in particular, onto $\{t_2^2, t_3^2, t_4^2, t_5^2\}$ through $Y_1 + \delta_1$, and onto $\{t_1^2, t_3^2, t_4^2, t_5^2\}$ through $Y_2 + \delta_2$.

Table 3: Noncentralities in $\mathcal{L}(t_i^2 | \delta_1) = t^2(1, \lambda_{11}, \lambda_{12})$ induced in succession by $\{Y_1 \rightarrow Y_1 + \delta_1\}$ in columns 2 and 3; and in $\mathcal{L}(t_i^2 | \delta_2) = t^2(1, \lambda_{21}, \lambda_{22})$ induced by $\{Y_2 \rightarrow Y_2 + \delta_2\}$ in columns 5 and 6; together with probabilities $\beta(\pm 2)$.

Item	$\{Y_1 \rightarrow Y_1 + \delta_1\}$			$\{Y_2 \rightarrow Y_2 + \delta_2\}$		
t_i^2	λ_{11}	λ_{12}	$\beta(\pm 2)$	λ_{21}	λ_{22}	$\beta(\pm 2)$
t_1^2	0.15278 δ_1^2	0.00000 δ_1^2	0.31450	0.58846 δ_2^2	0.03654 δ_2^2	0.43528
t_2^2	0.14385 δ_1^2	0.00893 δ_1^2	0.30622	0.62500 δ_2^2	0.00000 δ_2^2	0.47058
t_3^2	0.02778 δ_1^2	0.12500 δ_1^2	0.20998	0.25000 δ_2^2	0.37500 δ_2^2	0.18835
t_4^2	0.02838 δ_1^2	0.12440 δ_1^2	0.21043	0.02500 δ_2^2	0.60000 δ_2^2	0.08968
t_5^2	0.06187 δ_1^2	0.09091 δ_1^2	0.23614	0.11608 δ_2^2	0.50892 δ_2^2	0.12443

C_1 : Specifically, $\mathcal{L}(t_2^2 | \delta_1) = t^2(1, 0.143846 \delta_1^2, 0.008932 \delta_1^2)$ is stochastically smaller but near $\mathcal{L}(t_1^2 | \delta_1) = t^2(1, 0.152778 \delta_1^2, 0)$. From Definition 3 the *swamping probability*, that δ_1 swamps $\delta_2 = 0$, is $\beta_S(\delta_1) = P((t_2^2 | \delta_1) > c^*) = 0.30622$ at $\delta_1 = \pm 2$, in comparison with the power $\beta_P(\delta_1) = P((t_1^2 | \delta_1) > c^*) = 0.31450$ at $\delta_1 = \pm 2$ from Table 3.

C_2 : In consequence, $\delta_2 = 0$ is swamped by $\delta_1 = \pm 2$ with essentially the same probability as the power of t_1^2 to detect that Y_1 is in fact outlying. Similar comments apply for $\{Y_2 + \delta_2\}$, where the probability that δ_2 swamps $\delta_1 = 0$ is 0.43528 at $\delta_2 = \pm 2$, in comparison with the power 0.47058 at $\delta_2 = \pm 2$ from Table 3. Again δ_1 is swamped with likelihood approximating the power of t_2^2 to discern correctly that Y_2 is outlying.

C_3 : For $\{Y_2 + \delta_2\}$, evidence against $H_0: \delta_2 = 0$, in addition to t_2^2 , accrues through *small* t_4^2 and t_5^2 , neither outlying, by virtue of their considerably larger λ_2 than λ_1 . Specifically, we infer that t_4^2 , if coupled with a *lower* cutoff rule, *i.e.* t_4^{-2} against an upper cutoff value, would weigh in against $H_0: \delta_2 = 0$.

C_5 : This strange conclusion follows since $\mathcal{L}(t_4^{-2} | \delta_2) = t^2(1, 0.6000 \delta_2^2, 0.0250 \delta_2^2)$, its β probability being $\beta(\delta_2) = P((t_4^{-2} | \delta_2) > c^*) = 0.446230$ at $\delta_2 = \pm 2$. This in turn compares with the power $P((t_2^2 | \delta_2) > c^*) = 0.47058$ at $\delta_2 = \pm 2$ from Table 3, despite that Y_4 is *not* outlying. In particular, the power curve using $\{t_4^{-2} > c^*\}$ would approximate from below that of $\{t_2^2 > c^*\}$.

4.3. Pairs of outliers

If both (Y_1, Y_5) undergo nonvanishing shifts (δ_1, δ_5) , then for $\mathcal{L}(t_5^2)$ Theorem 2(i) gives

$$\lambda_{11} = 0.06186 \delta_1^2 - 0.19444 \delta_1 \delta_5 + 0.15278 \delta_5^2,$$

the denominator still at $\lambda_{12} = 0.09091 \delta_1^2$. Were (δ_1, δ_5) to be related linearly as $\delta_5 = 0.63634 \delta_1$, then $\mathcal{L}(t_5^2 | \delta_5 = 0.63634 \delta_1) = t^2(1, 0, 0.09091 \delta_1^2)$, categorically masking that both $\{Y_1 + \delta_1, Y_5 + \delta_5\}$ are outlying, however large in magnitude with $\delta_5 = 0.63634 \delta_1$. If instead $\delta_1 = -\delta_5$, then $\mathcal{L}(t_5^2 | \delta_1 = -\delta_5) = t^2(1, 0.40908 \delta_5^2, 0.09091 \delta_5^2)$ follows similarly. In comparison with $\mathcal{L}(t_5^2 | \delta_5^2) = t^2(1, 0.15278 \delta_5^2, 0)$ at $\delta_1 = 0$, the event $\delta_1 = -\delta_5$ serves to enhance the sample evidence against

$H_0: \delta_5 = 0$ using t_5^2 , in comparison with $\delta_1 = 0$. These examples serve to illustrate further prospects for irregularities to be incurred through pairs of outliers.

4.4. Inflation of residuals

For the *OLS* residuals and their moduli in Table 1, Table 4 lists expectations for $\mathcal{L}(e_i^2 | \delta_i)$ from

Table 4: Expectations for $\mathcal{L}(e_i^2 | \delta_i)$ and $\mathcal{L}(|e_i| | \delta_i)$, as induced in succession by $\{Y_1 + \delta_1\}$ and $\{Y_2 + \delta_2\}$ as in Table 3, with $\delta' = [\delta_1, \delta_2]$ evaluated at $\delta_1 = \delta_2 = \pm 2$, where $E(|e_i|)$ is to the order of approximation in (2).

Item	$[\delta_1, \delta_2] = [0, 0]$		$\{Y_1 \rightarrow Y_1 + \delta_1\}$		$\{Y_2 \rightarrow Y_2 + \delta_2\}$	
e_i^2	$E(e_i^2)$	$E(e_i)$	$E(e_i^2)$	$E(e_i)$	$E(e_i^2)$	$E(e_i)$
e_1^2	0.15278	0.29315	0.24615	0.38994	0.51240	0.62501
e_2^2	0.62500	0.59293	0.98462	0.77730	2.18750	1.29792
e_3^2	0.44445	0.50000	0.49383	0.52881	0.88890	0.76604
e_4^2	0.62500	0.59293	0.69594	0.62784	0.68750	0.62358
e_5^2	0.15278	0.29315	0.19059	0.33172	0.22372	0.36663

Theorem 1(i), and for $\mathcal{L}(|e_i| | \delta_i)$ from Theorem 1(ii), as induced by $\{Y_1 + \delta_1\}$ and $\{Y_2 + \delta_2\}$, and evaluated at $\delta_1 = \delta_2 = \pm 2$. Were there no shifts, $E(e_i^2) = (1 - h_{ii})$ and $E(|e_i|) \doteq 3(1 - h_{ii})^{1/2}/4$ follow from Theorem 1(i)(ii) as in columns 2 and 3. The extent of inflation owing to shifts is gaged by $[E(e_i^2) - (1 - h_{ii})]$ and $[E(|e_i|) - 3(1 - h_{ii})^{1/2}/4]$; these are often smaller for $\{Y_1 + \delta_1\}$ with leverage 0.84722, than for $\{Y_2 + \delta_2\}$ with leverage 0.37500.

4.5. Robust regression

Irregularities in residuals bear directly on *robust regression*, intended to mitigate the undue “pull” of shifts onto *OLS* regression through squared residuals. One approach seeks to solve $\sum_{i=1}^n \psi(\frac{e_i}{\hat{\sigma}}) \mathbf{x}_i = \mathbf{0}$, taking $\psi(\cdot)$ as an *influence function*. Specifically, with $e_i^* = e_i/\hat{\sigma}$, Huber’s *M*-estimator for β seeks to minimize $\sum_{i=1}^n \rho(e_i^*)$ as given by

$$\rho(e_i^*) = e_i^{*2}/2, \text{ for } |e_i^*| \leq r; \quad \rho(e_i^*) = r |e_i^*| - r^2/2, \text{ for } |e_i^*| > r \quad (6)$$

with r as a *tuning parameter*. Here $\hat{\sigma}$, often taken to be $\hat{\sigma} = 1.5 \text{med}\{|e_i|; 1 \leq i \leq n\}$, is intended as a *robust estimate for scale*. Nonlinearity leads to *Iteratively Reweighted Least Squares (IRLS)* supported by an algorithm in Myers (1990, p.351), for example. Findings here demand a critical reappraisal.

Table 4 under $\{Y_2 + \delta_2\}$ shows e_2^2 to be inflated stochastically and subject to down-weighting, as intended. Conversely, $\{e_1^2, e_3^2, e_4^2, e_5^2\}$, from unshifted observations, would appear to deserve full weights. To the contrary, each is inflated stochastically by virtue of $\{Y_2 + \delta_2\}$ as seen in Table 4 and, accordingly, will tend to be unduly penalized as excessive. These appear as unintended consequences not recognized in a considerable literature. Nor is it clear that $1.5 \text{med}\{|e_1|, |e_2|, |e_3|, |e_4|, |e_5|\}$

serves as a *robust estimate for scale*. To the contrary, it is inflated since all moduli in Table 4 are inflated stochastically, as induced vicariously through other shifts. Such anomalies, inadvertent and heretofore unrecognized, would appear to place in serious contention the claims conventionally advanced in support of *IRLS* for modulating excessive *OLS* residuals.

5. Case B: Hadi and Simonoff example

5.1. Overview

Hadi and Simonoff (1993) reported artificial data from the model $\{\mathbf{Y}_0 = \mathbf{X}_1 + \mathbf{X}_2 + \boldsymbol{\epsilon}_0\}$ with $n = 25$. Standard normal errors are assigned to observations $\{4, \dots, 25\}$, but deterministic shifts $\{Y_i = X_{i1} + X_{i2} + 4; 1 \leq i \leq 3\}$ stand *in lieu of* normal errors, intended to be difficult to find using conventional diagnostics. The authors advocate a two-step procedure for isolating outlying subsets; at Step One an option is to screen selected subsets using observed F_I values, as done in their study.

5.2. Single-case deletions

With regard to inherent masking, Table 5 is informative. Chiang (2007) simulated $N = 1000$ runs, and values for $\{t_i; 1 \leq i \leq 25\}$ were computed for each. The frequency with which all three outliers appeared in the top 20% of values $\{|t_i|; 1 \leq i \leq 25\}$, was empirically shown to be 82.5%. This serves to gage the overall extent of masking among three outliers as configured in that study. Tools developed here offer further insight.

In particular, we examine the manner in which shifts $[Y_1 + c, Y_2 + c, Y_3 + c]$ necessarily distribute as noncentralities across $\{\mathcal{L}(t_i^2) = t^2(22, \lambda_1, \lambda_2); 1 \leq i \leq 25\}$. The conventional test for $H_0: \delta = 0$ against $H_1: \delta \neq 0$, with null distribution $\mathcal{L}(t_i^2) = t^2(22, 0, 0)$ and critical value $c_\alpha^2 = 4.3009$ at level $\alpha = 0.05$, assumes a single outlier at a specified locus, and $\boldsymbol{\gamma} = \mathbf{0}$ at all other loci. However, the latter condition fails here since all of $\{Y_1, Y_2, Y_3\}$ are outlying by construction. Nonetheless, $\mathcal{L}(t_i^2) = t^2(22, 0, 0)$ continues to serve as reference for $\{\mathcal{L}(t_i^2); 1 \leq i \leq 25\}$ at c_α^2 , as in Remark 3. Table 5 reports noncentralities $\{(\lambda_1(c), \lambda_2(c))\}$ for all 25 cases, to include the probabilities $\beta(c) = P(t_i^2 > 4.3009)$ for $\{c = \pm 2, c = \pm 4\}$. If $\{Y_1 + c\}$ alone were outlying, then $\mathcal{L}(t_1^2) = t^2(22, 0.89515c^2, 0)$ since $(1 - h_{11}) = 0.89515$. Instead we see from Table 5 that $\mathcal{L}(t_1^2 | c[1, 1, 1]) = t^2(22, 0.52649c^2, 1.53616c^2)$ is stochastically diminished from this owing to additional outliers $\{Y_2 + c, Y_3 + c\}$. Further computations show that δ_1 is partially masked by (δ_2, δ_3) , the actual power at $c = \pm 4$ from $\mathcal{L}(t_1^2 | c[1, 1, 1]) = t^2(22, 0.52649c^2, 1.53616c^2)$ being only 49.257% of that, had $\{Y_1 + 4\}$ been the only outlier. This diminished power thwarts the capacity for t_1^2 to identify Y_1 to be the outlier that it is. Similar comments apply for (Y_2, Y_3) . These probabilities, diminished by masking, clearly factor into the empirical findings reported in Chiang (2007).

As swamping may be at issue, our tools again offer insight. A non-outlying $\{Y_j, j \geq 4\}$ is considered to be swamped by $[Y_1, Y_2, Y_3] + c[1, 1, 1]$ whenever $\{t_j^2 > c_\alpha^2\}$. Clearly for each $\{Y_j, j \geq 4\}$, the probabilities $\beta(c)$ in columns 5 and 6 of Table 5 are points along its *swamping curve*, namely, $\beta_S(c) = P(t_j^2 > c_\alpha^2 | c[1, 1, 1])$. For $\{j \geq 4\}$ it is noteworthy that $\beta(\pm 2) > \beta(\pm 4)$, as the denominator noncentralities become more assertive. Leverages here assume a pivotal role. Taking $\{h_{ii} \leq 0.1000\}$,

Table 5: The probabilities $\beta(c)$ for all single-case diagnostics $\{t_i^2; 1 \leq i \leq 25\}$ for the Hadi and Simonoff (1993) Table 1 data, as simulated instead in Chiang (2007). The value $c_\alpha = 4.3009$ at $\alpha = 0.05$.

t_i^2	h_{ii}	λ_1	λ_2	$\beta(\pm 2)$	$\beta(\pm 4)$
t_1^2	0.104848	0.52649 c^2	1.53616 c^2	0.20579	0.46661
t_2^2	0.104150	0.52768 c^2	1.53497 c^2	0.20628	0.46806
t_3^2	0.103455	0.52888 c^2	1.53377 c^2	0.20678	0.46952
t_4^2	0.138463	0.11686 c^2	1.94579 c^2	0.05560	0.04412
t_5^2	0.006250	0.00494 c^2	2.05771 c^2	0.02426	0.00333
t_6^2	0.093323	0.03350 c^2	2.02915 c^2	0.03176	0.00991
t_7^2	0.066544	0.02170 c^2	2.04095 c^2	0.02862	0.00690
t_8^2	0.022180	0.02104 c^2	2.04161 c^2	0.02844	0.00674
t_9^2	0.038713	0.02712 c^2	2.03553 c^2	0.03005	0.00823
t_{10}^2	0.047986	0.03287 c^2	2.02978 c^2	0.03159	0.00974
t_{11}^2	0.095735	0.01982 c^2	2.04283 c^2	0.02812	0.00645
t_{12}^2	0.017647	0.00075 c^2	2.06190 c^2	0.02320	0.00257
t_{13}^2	0.110563	0.08184 c^2	1.98081 c^2	0.04524	0.02693
t_{14}^2	0.044436	0.02183 c^2	2.04082 c^2	0.02865	0.00693
t_{15}^2	0.161999	0.02134 c^2	2.04131 c^2	0.02852	0.00681
t_{16}^2	0.009303	0.00699 c^2	2.05566 c^2	0.02479	0.00372
t_{17}^2	0.116990	0.09761 c^2	1.96504 c^2	0.04984	0.03416
t_{18}^2	0.148616	0.05552 c^2	2.00713 c^2	0.03777	0.01672
t_{19}^2	0.053639	0.00727 c^2	2.05538 c^2	0.02486	0.00378
t_{20}^2	0.191039	0.05209 c^2	2.01056 c^2	0.03682	0.01556
t_{21}^2	0.006125	0.00574 c^2	2.05691 c^2	0.02447	0.00348
t_{22}^2	0.000030	0.00003 c^2	2.06262 c^2	0.02301	0.00244
t_{23}^2	0.187475	0.02719 c^2	2.03546 c^2	0.03007	0.00825
t_{24}^2	0.096140	0.06689 c^2	1.99576 c^2	0.04096	0.02085
t_{25}^2	0.034350	0.00252 c^2	2.06013 c^2	0.02365	0.00289

the corresponding fifteen swamping probabilities in Table 5 are seen to satisfy $\{\beta_S(\pm 4) \leq 0.02085\}$. Moreover, a first-order Bonferroni bound gives a probability at least 0.90204 that none of the fifteen is swamped. Further assessments along these lines would be supported by Table 5.

6. Conclusions

The identification of *outlying* or *influential* observations, alleged to be “relatively simple” under single-case deletions, is reexamined here, our approach being probabilistic. For all diagnostics in $\mathfrak{D}$ in Appendix A, single and multiple shifts are seen to distribute as numerator and denominator noncentralities, all depending on regressors $\mathbf{X}_0$ independently of responses $\mathbf{Y}_0$. An informative di-

chotomy results on projecting shifts into the “Regressor” and “Error” spaces of the reduced model. A further finding is that squared *OLS* residuals $\{e_i^2 = (Y_i - \hat{Y}_i)^2; 1 \leq i \leq n\}$ may be inflated or deflated stochastically, even when a given Y_j remains unshifted. Constructs akin to power curves track probabilities of *swamping* and *masking* of one shift by others as shifts evolve. Here $\mathbf{X}_0$ is fixed; if random, our findings apply conditionally in keeping with the *conditionality principle*. Accordingly, buried within each design are seeds for these aberrant phenomena, as are drawn out and articulated here.

These facts now place in disarray the meanings to be ascribed to single-case deletion diagnostics. In addition, the stochastic inflation of e_j^2 , even when Y_j is not outlying, casts into question the *IRLS* algorithm as a palliative to *OLS* against excessive residuals. In short, despite wide acceptance, supporting software, and routine usage, these procedures fail to achieve essential objectives traditionally sought. A Referee states: “It is unclear what is the point of noting that irregularities may appear in a situation when 3 out of 5 observations are shifted,” as in Section 4. Reply: This illustrates in the extreme, for an elementary and transparent model, not only that unfettered shifts are allowed, but more importantly, that our tools are equipped to evaluate their consequences.

To place this study in perspective, we have examined properties of ordinary residuals and their conjuncts under *OLS*, as understood in many disciplines and supported by software packages. As demonstrated, we secure exact distributional results under conventional assumptions, but allowing for shifts at various loci. Proponents for *robust regression* instead might prefer alternatives to *OLS* other than *IRLS* at the outset. Such criteria not only are computer intensive, but the unavailability of analytical results remands the available tools to simulative, as noted in Hadi and Simonoff (1993). Moreover, our models postulate neither “clean” nor “contaminated” subsets, so that the notion of *high-breakdown* estimators, referring to the fraction of contaminants allowed, is vacuous. Even for the data of Section 5, where $\{Y_j; j \geq 4\}$ remain unshifted, thus “clean,” simulation studies in Hadi and Simonoff (1993) show no superiority, over *OLS*, of the criterion *Least Median of Squared Residuals*, as favored by some.

Appendix A – Diagnostics

Conventional single-case deletion diagnostics include

$$\mathfrak{D} = \{t_i, DFFITs, DFBETAs, COVRATIOs, FVARATIOs, OUT_i, AP_i, C_i, WK_i, W_i, D_i\}; \quad (7)$$

the second through fifth from Belsley et al. (1980); OUT_i in Barnett and Lewis (1994); AP_i from Andrews and Pregibon (1978) C_i as in Cook (1977); WK_i from Welsch and Kuh (1977); W_i as in Welsch (1982); and with $\mathbf{V}^-$ as a reflexive g -inverse of $\mathbf{V}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_i)$, the diagnostic $D_i = (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_i)' \mathbf{V}^- (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_i) / S_i^2$ as in Jensen (2001). For detailed properties see Table 1 of Jensen (2000). Observations are deemed *influential*, whose removal would alter signal features of the analysis. Taking $DFFIT_i = (\hat{Y}_i - \hat{Y}_{i(i)}) / (S_i \sqrt{h_{ii}})$ as the scaled divergence between predictors at $\mathbf{x}_i$ with and without Y_i , the observation Y_i is deemed *influential for prediction* at $\mathbf{x}_i$ when its removal would alter $DFFIT_i$ sufficiently; and similarly for other influence diagnostics. Anomalies in these are certified in Chatterjee and Hadi (1986) and discussants; these essentially were resolved by the finding that all diagnostics in $\mathfrak{D}$ correspond one-to-one with t_i or t_i^2 ; Lamotte (1999) and Jensen (2000); see also Naik (2003). Accordingly, irregularities induced through outliers follow as in Corollary 1.

Acknowledgement

Thanks are extended to (i) Professor Donald E. Ramirez for helpful exchanges on these topics, and for computation of the β probabilities for the nonstandard distributions cited; and (ii) to an anonymous Referee for several helpful suggestions.

References

- ANDREWS, D. F. AND PREGIBON, D. (1978). Finding outliers that matter. *J. Roy. Statist. Soc. B*, **40**, 85–93.
- ATKINSON, A. C. (1985). *Plots, Transformations, and Regression*. Oxford UK: Oxford University Press.
- BARNETT, V. AND LEWIS, T. (1994). *Outliers In Statistical Data*. Second edition. New York: Wiley.
- BECKMAN, R. J. AND TRUSSELL, H. J. (1974). The distribution of an arbitrary studentized residual and the effects of updating in multiple regression. *J. Am. Statist. Assoc.*, **69**, 199–201.
- BELSLEY, D. A., KUH, E., AND WELSCH, R. E. (1980). *Regression Diagnostics: Identifying Influential Data And Sources Of Collinearity*. New York: Wiley.
- BOX, G. E. P. AND WILSON, K. B. (1951). On the experimental attainment of optimum conditions. *J. Roy. Statist. Soc. B*, **13**, 1–45.
- CHATTERJEE, S. AND HADI, A. S. (1986). Influential observations, high leverage points, and outliers in linear regression. *Statist. Sci.*, **1**, 379–393.
- CHATTERJEE, S. AND HADI, A. S. (1988). *Sensitivity Analysis In Linear Regression*. New York: Wiley.
- CHIANG, J. T. (2007). The masking and swamping effects using the planted mean – Shift outliers models. *Int. J. Contemp. Math. Sciences*, **2**, 297–307.
- COOK, R. D. (1977). Detection of influential observations in linear regression. *Technometrics*, **19**, 15–18.
- COOK, R. D. AND WEISBERG, S. (1982). *Residuals And Influence In Regression*. London: Chapman & Hall.
- FOX, J. (1991). *Regression Diagnostics*. Newbury Park, CA: Sage.
- GENTLEMAN, J. F. AND WILK, M. B. (1975). Detecting outliers. II. Supplementing the direct analysis of residuals. *Biometrics*, **31**, 387–410.
- GHOSH, S. (1978). On robustness of designs against incomplete data. *Sankhyā*, **40**, 204–208.
- HADI, A. S. AND SIMONOFF, J. S. (1993). Procedures for the identification of multiple outliers in linear models. *J. Am. Stat. Assoc.*, **88**, 1264–1272.
- HOAGLIN, D. C. AND KEMPTHORNE, P. J. (1986). Influential observations, high leverage points, and outliers in linear regression: Comment. *Statist. Sci.*, **1**, 408–412.
- HOCKING, R. R. AND PENDLETON, O. J. (1983). The regression dilemma. *Commun. Statist. Theor. Meth.*, **12**, 497–527.
- JENSEN, D. R. (2000). The use of studentized diagnostics in regression. *Metrika*, **52**, 213–223.
- JENSEN, D. R. (2001). Properties of selected subset diagnostics in regression. *Statist. Probab. Letters*, **51**, 377–388.

- JENSEN, D. R. (2010). Leverage and subset efficiencies in regression. *Statist. Method.*, **7**, 541–551.
- JENSEN, D. R. AND RAMIREZ, D. E. (2014). Noncentralities induced in regression diagnostics. *J. Statist. Theory & Practice.*, **8**, 141–165.
- LAMOTTE, L. R. (1999). Collapsibility hypotheses and diagnostic bounds in regression analysis. *Metrika*, **50**, 109–119.
- MYERS, R. H. (1990). *Classical And Modern Regression With Applications*. Second edition. Boston, Ma: Pws-Kent.
- NAIK, D. N. (2003). Diagnostic methods for univariate and multivariate normal data. In KHATTREE, R. AND RAO, C. R. (Editors) *Handbook Of Statistics 22: Statistics In Industry*. Amsterdam: Elsevier Science B.V., pp. 957–993.
- NANDI, S. B. AND CHOUDHURY, S. (2002). Series representation of doubly noncentral t distribution by use of the Mellin integral transform. *Commun. Statist. Theor. Meth.*, **31**, 2139–2149.
- ROUSSEEUW, P. J. AND LEROY, A. M. (1987). *Robust Regression And Outlier Detection*. New York: Wiley.
- ROUSSEEUW, P. J. AND VAN ZOMEREN, B. C. (1990). Unmasking multivariate outliers and leverage points. *J. Am. Statist. Assoc.*, **85**, 633–639.
- SNEDECOR, G. W. AND COCHRAN, W. G. (1968). *Statistical Methods*. Sixth edition. Ames, IA: Iowa State University Press.
- WELSCH, R. E. (1982). Influence functions and regression diagnostics. In LAUNER, R. L. AND SIEGEL, A. F. (Editors) *Modern Data Analysis*. New York: Academic Press.
- WELSCH, R. E. (1983). Leverage. In JOHNSON, N. L., KOTZ, S., AND READ, C. B. (Editors) *Encyclopedia Of Statistical Sciences*, volume 4. New York: Wiley, pp. 610–611.
- WELSCH, R. E. AND KUH, E. (1977). Linear regression diagnostics. Technical report, Working Paper 923–77 Sloan School Of Management, Cambridge, MA: MIT.